Smriti Singh

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Summary

Computer Science undergraduate specializing in AI & Machine Learning with experience building end-to-end AI applications using Python, LangChain, Google Gemini, Scikit-learn, Flask, and Streamlit. Strong foundation in software engineering, machine learning, REST APIs, and cloud deployment.

Technical Skills

Languages: Python, C++, Java, JavaScript, SQL

AI/ML: LangChain, Google Gemini API, Scikit-learn, TensorFlow, PyTorch, Predictive Modeling, Feature Engineering

Backend: Flask, Node.js, REST APIs

Frontend: Streamlit, HTML, CSS, JavaScript

Databases: MongoDB, MySQL

Libraries & Tools: Pandas, NumPy, BeautifulSoup, Git, GitHub, Docker, Render, MLflow

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Projects

Multi-Agent AI Research Assistant — Jun 2026 – Jul 2026

Live Demo | GitHub — *Python, LangChain, Google Gemini, Streamlit, Tavily Search API, BeautifulSoup, Render*

- Architected and deployed a multi-agent AI system using LangChain and Google Gemini to automate web research, structured report generation, and AI-driven report evaluation.
- Built custom LangChain tools integrating Tavily Search API and BeautifulSoup for real-time web search, content extraction, and multi-source information synthesis.
- Implemented Search, Writer, and Critic agents using prompt engineering, tool calling, and agent orchestration to generate comprehensive research reports with cited sources.
- Deployed the application on Render with a Streamlit frontend, enabling users to generate AI-powered research reports and review AI-generated feedback through an interactive interface.

Customer Churn Prediction System — May 2026 – Present

Live Demo — *Python, Scikit-learn, Streamlit, Docker, Flask, Render*

- Built and deployed an end-to-end machine learning application for customer churn prediction using Scikit-learn, Streamlit, Docker, and Render.
- Developed a complete ML pipeline including preprocessing, feature engineering, model training, evaluation, and real-time inference.
- Implemented Logistic Regression achieving **82% accuracy** and **0.86 ROC-AUC** for customer churn classification.
- Designed an interactive dashboard providing churn probability, customer risk scoring, and actionable retention insights.
- Containerized the application using Docker and deployed a production-ready inference service.

Crop Recommendation System — Feb 2025 – Apr 2025

Full-Stack ML Project — *Python, Flask, Node.js, MongoDB, OpenWeather API*

- Developed a machine learning-powered crop recommendation system using soil nutrients, rainfall, and climate features for personalized recommendations.
- Built RESTful APIs with Flask and Node.js to serve prediction requests and process real-time environmental data.
- Integrated the OpenWeather API to generate location-aware crop recommendations using live weather conditions.
- Designed a responsive frontend enabling users to receive personalized recommendations through an intuitive interface.

Education

- **VIT Bhopal University, Bhopal** — Sep 2023 – Present
  Bachelor of Technology in Computer Science (AI & ML Specialization)
  CGPA: 9.22
- **Vidya Bharati Chinmaya Vidyalaya, Jamshedpur** — 2020 – 2022
  12th Grade: 86.8% (2022)
- **Hill Top School, Jamshedpur** — 2014 – 2020
  10th Grade: 94% (2020)

Certifications

- Microsoft Azure Data Fundamentals Certified
- Machine Learning in Collaboration with Google for Developers